

Staff level AI backend infra engineer. 6 years of professional experience. \$10M business impact.

Cross-disciplinary technical breadth in Software and Hardware Engineering, ML, Security, Product, and Infra

Experienced people leader, fun to work with, appreciate smart engineers with low ego. Friendly, honest, and genuine

Conducted dozens of algorithm and system design interviews

Proven record in strategize and deliver reliable systems at scale through cross-team collaborations

WORK EXPERIENCE

Samaya AI

London

Member of Technical Staff

Jan 2024 - Current

- Architect Knowledge Ingestion Pipeline, Embedding Chunking strategy, Eval, RAG, and AI Infrastructure
- 0-1 build product features like DeepResearch, WorkflowAgents, QueryDecomposition, StockPicker
- Decrease time-to-first-token from 100 to 10 seconds (-90%), scale infra from 10 to 10k daily active user
- Onboard customers from BlackRock, Citi, and Point72. Strategize pricing. Demo for our \$43.5M series A

Tesla

Shanghai GigaFactory

High Voltage Manufacture Org

Texas GigaFactory

Senior Software Engineer (manage a team of 7 people)

July 2022 – Jan 2024

- Responsible for high voltage system for Tesla Model 3, Y, S, X, Powerwall, Megapack, and SuperCharger.
- Oversee development on charger, tester, network conversion, eFuse, PLC controller, and C++ firmware
- Iterate chip design for CyberTruck and Megapack under tight deadline (2.5 months) and heavy pressure
- Create battery cell simulation for easier debugging and testing. Decouple software and hardware dev loop
- Design new factory software framework, pipeline, dashboards, from lab to production, Berlin to Palo Alto
- Close collaboration with worldwide vendors including CATL, BYD, TI, COSMX, HYC, Pegatron, and Jabil.

Often travel to their countries for design review, quality assurance, troubleshooting, and inspections

- Answer questions and teach Manufacture, Process, and Quality teams in factory to enable them to do their own debugging and fault analysis. This helped making China battery production line 3x faster than USA line
- Manage a team of 2 technicians, 2 software, 1 mechanical, and 1 electrical engineer
- Finish CyberTruck battery charger 4 weeks ahead of schedule
- Bring up 50 new manufacture stations, maintain over 200 stations, handle \$500k budget per year

Atlassian

Mountain View, CA

Cloud Security + Platform Infrastructure

Fully remote

Software Engineer

2020 – July 2022

- Develop auth module, required for all communication between Atlassian microservices and staff
- Rewrite tools like Idap cache, rate limiter, and deduper from Java, Typescript, and Python to Golang
- On call and answer questions for a 99.99% availability internal PaaS hosting over 1400 of Atlassian services
- Create instant generator for services with bells and whistles like k8s, SOX compliant, and CI/CD
- Tools: Splunk, SignalFX, Jira, Confluence, AWS S3, Dynamodb, Lambda, Postgres, Spinnaker

TECH STACK

Golang, Python, C++

GCP, AWS, k8s, gRPC, GraphQL, vLLM, Grafana, Opentelemetry, ElasticSearch, MongoDB, Postgres, Cursor

Strong desire to learn new tech at work

EDUCATION

Johns Hopkins University

Master of Science GPA: 3.9/4.0

2019

Information Security

Courses: Cloud Security, Software Vulnerability Analysis, Security Analytics, Computer Forensics,
Network security, Privacy Laws, Risk Management, Ethical hacking, Modern Cryptography

University of Rochester

Bachelor of Science GPA: 3.6/4.0

Joseph C Wilson "Change" Scholarship; Dean's list

Electrical Engineering and Computer Engineering

2014

Focus: signal processing, robot control, semiconductor, microprocessor, C++, Java, assembly, VHDL, MATLAB.

Minor: Mathematics and Economy

SIDE PROJECTS

Agemo

remote

- Build a fullstack webapp for displaying execution history with NodeJS, and Shadcn for codewords.ai

Rippling

remote

- Build out global expansion framework for Rippling's payroll product using PythonDjango+MongoDB+React

Unknot.id

Baltimore, MD

Security Researcher Intern

Aug-Dec 2019

- Implement black-box adversarial attacks against a behavior biometric auth mechanism on Android
- Harden the LSTM model robust against this <1000 queries shadow model attack building upon CleverHans

Cognitive Modeling of User Security

Baltimore, MD

Researcher at JHU Computer Science Department

Jan-Aug 2019

- Build computational cognitive modeling to simulate human decision-making in Python
- Adapt Instance-Based Learning Theory and ACT-R memory chunk activation and analyze 40GB data

Drone System

Rochester, NY

Developer

Sep-Dec 2017

- Create a drone drop system for National Training Center and Joint Readiness Training Center of U.S. Army
- 3D CAD Design and print remote controller, release mechanism, and program microprocessors in Arduino

Entrepreneurship

- Cheap long distance hitchhiking platform

December 2017

- Digital library for mixing drinks app

May 2017

- Language exchange website to help people find English/Chinese speaking partner

April 2015

Docker privilege escalation

- Research and apply Dirty COW race condition to escape from container to host

IoT Exploit

- Scan, discover, and capture network communications from DJI drones, Bebop drones, and Amazon Echo
- Look for vulnerabilities and execute attacks to successfully disable the above devices

Malicious Website Detection through URL Analytics

- Gather dataset from PhishTank for malicious URLs and DMOZ for benign URLs
- Utilize machine learning models in JupyterNotebook and use ROC curve to measure performances

Software Vulnerability Analysis

- Decompose and create threat models for open source software at a high-level
- Reverse engineering the programs with objdump, ldd, nm, and strace

CryptoDoneRight

- Assist in building cryptodoneright.org (a training tool for server hardening), and test its effectiveness

Leetcode

- Solved 1000+ questions, improve contest rating from 1460 to 2109 in 7 months

OTHER ACTIVITIES

Gaming Dev	https://battle-map-344101.wl.r.appspot.com/
Entrepreneurship	hitchhiking platform; Echinglish.weebly.com
DIY electronics	headphone amplifier; windshield wiper controller; pc
Music	guitar player; UofR carillon society carillonneur. Multiple live concert performance
Sports	standup paddle board enthusiast, top 0.3% LoL player, amateur boxer, 10 pull ups

URLs:

<https://zuo.vercel.app/>

<https://www.linkedin.com/in/mrmightymango/>

<https://leetcode.com/mangoman/>

<https://github.com/MrMegaMango>

<http://behavior.isi.jhu.edu/people.html>

<https://echinglish.weebly.com/>

<https://prezi.com/view/bhBhNzij1gost2k0CTnM/>

<https://www.unknot.id/post/yes-you-can-spoof-behavioral-biometrics-with-adversarial-learning>